

1 Sup Inf & Set || Nested Intervals || Sequence & Bolzano Weierstrass

1.1 Supremum and Infimum & Set

Completeness Axiom: If $\exists$ UP $\Rightarrow$ LUB (sup) ; If $\exists$ LB $\Rightarrow$ GLB (inf). **Archimedean Property:** $\forall x \in \mathbb{R}^+ \Rightarrow \exists n \in \mathbb{N} \text{ s.t. } n > x$.

Countable Set: A set E is countable if $\exists$ bijection $f: \mathbb{N} \rightarrow E$. ps: $[0, 1]$ is uncountable.

· **Property:** Let $(E_n)_{n \in \mathbb{N}}$ be countable sets. $\Rightarrow$ ¹ $\cup E_n$ is countable. (Can infinite) ² $\prod E_n$ is countable. (Finite) ³ $[0, 1]$ uncountable.

[⊕] **Dense Set:** Set A is dense in $\mathbb{R}$ iff $\forall x \in \mathbb{R}, \forall \varepsilon > 0 \exists a \in A \text{ s.t. } |x - a| < \varepsilon$ e.g. [⊕] $\mathbb{Q}$ is dense in $\mathbb{R}$

[⊕] **Power Set $\mathcal{P}$:** $\mathcal{P}(E)$ is the set of all subsets of E . ps: $\mathcal{P}(\mathbb{R})$ is the set of all subsets of $\mathbb{R}$.

Property of Sup, Inf: $\inf(-f_n) = -\sup(f_n) \parallel \limsup(-f_n) = -\liminf(f_n) \parallel$ If g is fixed, $\sup(f_n + g) = \sup(f_n) + g$ ps: 对 inf, lim sup, lim inf 也成立

1.2 Nested Intervals

Def of $\lambda(I)$: Let I be an interval. $\lambda(I) := \sup I - \inf I \parallel \lambda((a, b)) = \lambda([a, b)) = \lambda((a, b]) = \lambda([a, b]) = b - a$.

Nested Intervals: Sequence $(I_n)_{n \in \mathbb{N}}$ is nested if $I_n \supseteq I_{n+1} (\downarrow)$ for all $n \in \mathbb{N}$.

Nested Interval Theorem: If $(I_n)_{n \in \mathbb{N}}$ is nested closed bounded interval sequence $\Rightarrow E := \cap I_n \neq \emptyset$.

· **property:** ¹ If $\lim \lambda(I_n) = 0 \Rightarrow E = a$ ² [⊕] If $I_n = [a_n, b_n] \Rightarrow E = [\sup a_n, \inf b_n]$

1.3 Sequence, Cauchy & Bolzano Weierstrass Theorem

Useful Seq Limit: $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0, p > 1$; $\lim a^n = 0, |a| < 1$; $\lim n^{\frac{1}{n}} = 1$; $\lim a^{\frac{1}{n}} = 1, a > 0$

Operation in Seq Limit: Squeeze Theorem ; Monotone Convergence Theorem (MCT) ; $\lim ax_n = a \lim x_n$

· $\lim(x_n + y_n) = \lim x_n + \lim y_n$; $\lim x_n \cdot y_n = \lim x_n \cdot \lim y_n$; $\lim \frac{x_n}{y_n} = \frac{\lim x_n}{\lim y_n}, y_n \neq 0$

Comparison Theorem (For Seq): If $\lim x_n, y_n$ exist, and $x_n \leq y_n$ for all $n \geq N$ then $\lim x_n \leq \lim y_n$.

Cauchy Sequence: Sequence $(a_n)_{n \in \mathbb{N}}$ is Cauchy iff $\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } |a_n - a_m| < \varepsilon$ for all $m, n \geq N$. $\parallel$ **Convergent $\Leftrightarrow$ Cauchy**

Bolzano-Weierstrass Theorem: Every bounded sequence has a convergent subsequence.

Def of limit superior and inferior: $\limsup a_n := \lim(\sup_{k \geq n} a_k) \parallel \liminf a_n := \lim(\inf_{k \geq n} a_k)$

· **Property:** If $(a_n)_{n \in \mathbb{N}}$ bounded. $\Rightarrow$ [⊕] ¹ $\exists (a_{n_k}) \rightarrow \limsup a_n$ Hint: $\exists a_{n_k}, \sup_{j \geq k} a_j - \frac{1}{k} \leq a_{n_k} \leq \sup_{j \geq k} a_j, \sup_{j \geq n_k} a_j \leq \sup_{j \geq k} a_j$ [⊕] ² $\exists (a_{n_k}) \rightarrow \liminf a_n$
[⊕] ³ $\forall a_{n_k} \rightarrow a \liminf a_n \leq a \leq \limsup a_n$

Subsequence Theorem: In each case, the subsequence can be taken to be *monotonic*.

1. $\exists a \in \mathbb{R} \text{ s.t. } \forall \varepsilon > 0$, there are infinitely many $n \in \mathbb{N} \text{ s.t. } |x_n - a| < \varepsilon$ ($\exists$ 聚点 a) $\Leftrightarrow \exists$ subsequence of $(x_n) \text{ s.t. } \lim x_{n_k} = a$
2. $(x_n)_{n \in \mathbb{N}}$ is not bounded above (below) $\Leftrightarrow \exists$ subsequence of $x_n \text{ s.t. } \lim x_{n_k} = \infty (-\infty)$
3. If $\lim x_n = L \Rightarrow \forall x_{n_k}, \lim x_{n_k} = L$

2 Series

Cauchy Criterion for Series: $\sum_{k=1}^{\infty} a_k$ converges $\Leftrightarrow \forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq m \geq N |\sum_{k=m+1}^n a_k| = |a_{m+1} + \dots + a_n| < \varepsilon$.

p-Test: $\sum \frac{1}{n^p}$ convergent iff $p > 1$ **Root Test** **Ratio Test** $|a_n|^{1/n}, |a_{n+1}|^{1/n+1}$ **Geometric Series:** $\sum p^n$ converges iff $|p| < 1$

Divergence Test: If $\lim a_n \neq 0 \Rightarrow \sum a_n$ diverges. (If $\sum a_n$ convergent $\Rightarrow \lim a_n = 0$.)

Comparison Test: If $0 < a_n < b_n, \sum b_n$ convergent $\Rightarrow \sum a_n$ also ; $\sum a_n$ divergent $\Rightarrow \sum b_n$ also.

· **Limit Comparison Test:** If $a_n, b_n \geq 0, \lim \frac{a_n}{b_n} = L > 0 \Rightarrow$ If $L \in (0, \infty)$: $\sum a_n$ convergent iff $\sum b_n$ also.

If $L = 0$: $\sum b_n$ convergent $\Rightarrow \sum a_n$ also. $L = \infty$: $\sum b_n$ divergent $\Rightarrow \sum a_n$ also.

Integral Test: Let $f: [1, \infty) \rightarrow \mathbb{R}$ is 非负递减, $a_n = f(n)$. Then $\sum a_n$ converges iff $\int_1^{\infty} f(x)dx < \infty$.

Absolutely Convergence: $\sum a_n$ convergent absolutely iff $\sum |a_n|$ convergent. **If convergent abs $\Rightarrow$ convergent.**

Alternating Series Test: If a_n decreasing, $a_n \geq 0, \lim a_n = 0$. Then $\sum (-1)^{n-1} a_n$ convergent.

Cauchy's Condensation Test: If $a_n \geq 0, a_n$ decreasing, $\Rightarrow [\sum a_n \text{ convergent} \Leftrightarrow \sum 2^n a_{2^n} \text{ also}]$

Riemann Rearrangement Theorem: If $\sum_{k=1}^{\infty} a_k$ is conditionally convergent. $\Rightarrow \forall s \in \mathbb{R}, \exists$ bijection $\sigma: \mathbb{N} \rightarrow \mathbb{N} \text{ s.t. } \sum_{n=1}^{\infty} a_{\sigma(n)} = s$.

Rearrangement for Abs Series: If $\sum_{n=1}^{\infty} a_n$ is absolutely convergent. $\Rightarrow \forall$ bijection $\sigma: \mathbb{N} \rightarrow \mathbb{N}, \sum_{n=1}^{\infty} a_{\sigma(n)} = \sum_{n=1}^{\infty} a_n$.

3 Continuous & Differentiable Functions Properties

Def of lim of func: $1. \forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } 0 < |x - x_0| < \delta, \Rightarrow |f(x) - L| < \varepsilon. \quad 2. \forall x_n \neq x_0 \text{ s.t. } \lim(x_n) = x_0, \text{ then } \lim f(x_n) = L$.

Def of cont of func: $1. \forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } \forall x, |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon. \quad 2. \forall (x_n)_{n \in \mathbb{N}}, \text{ if } \lim x_n = x_0 \Rightarrow \lim f(x_n) = f(x_0)$

Def of diffi: $1. f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \quad 2. f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$.

The Extreme Value Theorem (EVT): If $f: [a, b] \rightarrow \mathbb{R}$ is continuous, then f has a max and min value.

Intermediate Value Theorem (IVT): $f: [a, b] \rightarrow \mathbb{R}$ is continuo, $f(a) \neq f(b), y \in [f(a), f(b)] \Rightarrow \exists c \in (a, b) \text{ s.t. } f(c) = y$

IVT (for deriv) / (Darboux Theorem): If f differentiable in $I \Rightarrow \forall m \in (f'(a), f'(b))$ we have: $\exists c \text{ s.t. } f'(c) = m$

Rolle's Theorem: If f continuous on $[a, b]$, differentiable on (a, b) and $f(a) = f(b)$ Then: $\exists c \in (a, b) \text{ s.t. } f'(c) = 0$

Mean Value Theorem (MVT): Let f, g continuous on $[a, b]$ and differentiable on (a, b) : ps: $a=b$ 时依然成立

· $1. \exists c \in (a, b) \text{ s.t. } \frac{f(b) - f(a)}{b - a} = f'(c) \quad 2. \exists c \in (a, b) \text{ s.t. } \frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$

Theorem: If f is 1-1 and continuous $\Rightarrow f$ is strictly monotone. and f^{-1} is continuous and strictly monotone.

Inverse Function Theorem: Let f differentiable in I (open interval), and $\forall x, f'(x) \neq 0$:

· **1.** $f(I)$ is open interval. **2.** f is invertible. **3.** f^{-1} is differentiable and $(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$

Second Deriv Test: f twice continuously differentiable in $I, f'(x_0) = 0$: **1.** $f''(x_0) > 0 (< 0)$ local min (max) point.

4 Pointwise & Uniformly Convergence

4.1 Pointwise Convergence

Pointwise Convergence: Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of functions. $f_n : E \rightarrow \mathbb{R}$ converge pointwise to $f : E \rightarrow \mathbb{R}$ if: 不加说明默认 pointwise

1. **Version 1:** $\forall x \in E, \forall \varepsilon > 0, \exists N_{\varepsilon, x} \text{ s.t. } \forall n > N |f_n(x) - f(x)| < \varepsilon$

2. **Version 2:** $\forall x \in E, \lim f_n(x)$ exists and $\lim f_n(x) = f(x)$

Pointwise Convergence NOT Preserve: ¹continuity, ²differentiability, ³ $(\lim f_n)' \neq \lim f_n'$, ⁴ $\int \lim f_n \neq \lim \int f_n$

· 对应反例: $f_n : [0, 1] \rightarrow \mathbb{R} := x^n$ ² $f_n : (-1, 1) \rightarrow \mathbb{R} := \sqrt{x^2 + \frac{1}{n}}$ ³ $f_n : (-1, 1) \rightarrow \mathbb{R} := \frac{\sin(nx)}{n}$ ⁴ $f_n : (0, 1] \rightarrow \mathbb{R} := \begin{cases} n, & 0 < x \leq \frac{1}{n} \\ 0, & \frac{1}{n} < x \leq 1 \end{cases}$

4.2 Uniformly Convergence

Graph: $\Gamma(f) := \{(x, f(x)) | x \in E\}$ ps: $f : E \rightarrow \mathbb{R}$

Vertical Neighbourhood $\mathcal{N}(f, \varepsilon) := \{(x, y) | x \in E, |y - f(x)| < \varepsilon\}$ ps: $f : E \rightarrow \mathbb{R}$

Uniformly Convergence: $f_n : E \rightarrow \mathbb{R}$ converge uniformly iff: || **Uniformly $\Rightarrow$ Pointwise**

1. **Version 1:** $\forall \varepsilon > 0, \exists N_\varepsilon \text{ s.t. } \forall n > N, \forall x \in E, |f_n(x) - f(x)| < \varepsilon$

2. **Version 2:** $\lim [\sup_{x \in E} |f_n(x) - f(x)|] = 0$

3. **Version 3 | Uniformly Cauchy:** $\forall \varepsilon > 0, \exists N_\varepsilon \in \mathbb{N} \text{ s.t. } \forall x \in E, m, n > N |f_n(x) - f_m(x)| < \varepsilon$ || **Converge Unif $\Leftrightarrow$ Unif Cauchy**

4. **Version 4 | Vertical Neighbourhood Test:** $\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall n > N, \Gamma(f_n) \subset \mathcal{N}(f, \varepsilon)$

Sequence Test: If $f_n : E \rightarrow \mathbb{R}$ converge uniformly to f . $\Rightarrow$ If $(x_n)_{n \in \mathbb{N}} \in E, \lim |f_n(x_n) - f(x_n)| = 0$ 不能当作 $\forall x_n$

Properties of Uniformly Convergent: Let $f_n, f, g : I \rightarrow \mathbb{R}$ Then: NOT preserve: differentiable.

1. **Continuous:** If ¹ $\lim f_n(x) = f(x)$ uniformly, ² $\forall f_n$ continuous at $x_0 \Rightarrow f$ continuous at x_0 . $\Downarrow \forall x \in I, (\lim f_n)'(x) = \lim f_n'(x)$

2. **Termwise differentiation:** If ¹ I is bounded open; ² $\forall f_n$ diff, $\lim f_n' = g$ unif; ³ $\exists x_0 \in I \text{ s.t. } f_n(x_0)$ converges. $\Rightarrow \lim f_n = f$ unif; f' diff and $f' = g$.

3. **Uniform Convergence Theorem:** If ¹ $\lim f_n(x) = f(x)$ uniformly on $E \in \mathcal{M}, m(E) < \infty, {}^2 f_n \in \mathcal{L}^1(\mathbb{R}) \Rightarrow {}^1 f \in \mathcal{L}^1(E) {}^2 \lim \int_E f_n(x) dx = \int_E f dx$

4. ^④ If $f_n : [a, b] \rightarrow \mathbb{R}, g_n : [a, b] \rightarrow \mathbb{R}$ converge uniformly. $\Rightarrow \frac{f_n}{g_n}$ converge uniformly also. ps: (a, b) 不行 Hint: $f_n g - f g_n = f_n g - f g + f g - f g_n$

Def of Series of Functions: Let $f_n : E \rightarrow \mathbb{R}$; Def partial sum $s_n(x) := \sum_{k=1}^n f_k(x)$

1. $\sum f_n$ converge pointwise iff s_n converge pointwise. 2. $\sum f_n$ converge uniformly iff s_n converge uniformly. 3. $\sum f_n$ converge absolutely (pointwise) iff $\sum |f_n|$ converge (pointwise).

Weierstrass M-Test: $f_n : E \rightarrow \mathbb{R}, \exists (M_n)_{n \in \mathbb{N}} \geq 0 \text{ s.t. } \forall x \in E, \forall n \in \mathbb{N} : |f_n| < M_n; {}^2 \sum M_n$ converges. $\Rightarrow \sum f_n$ converge absolutely & Uniformly.

5 Power Series

5.1 Power Series, Radius of Convergence, Continuity and Differentiability

Power Series: A series from $\sum_{n=0}^{\infty} a_n(x-c)^n$ **Center:** c . **Coefficients:** a_n .

Radius of Convergence (R): For a power series $\sum a_n(x-c)^n$, there are three situations for it's radius of convergence:

1. $R = 0$ if $\sum a_n(x-c)^n$ converges only $x = c$.
2. $R = \infty$ if $\sum a_n(x-c)^n$ converges absolutely $\forall x \in \mathbb{R}$.
3. $\exists R > 0$ s.t. $\sum a_n(x-c)^n$ converges absolutely if $|x-c| < R$; diverges if $|x-c| > R$.
4. **All-In-One:** $R = \sup\{r \geq 0 : (a_n r^n)_{n \in \mathbb{N}} \text{ is bounded}\}$

Properties | In Radius of Convergence: Assume $f(x) = \sum a_n(x-c)^n$ with $R > 0$.

1. **Continuous:** $f(x)$ continuous on $(c-R, c+R)$.
2. **Convergence:** $\forall 0 < r < R, f(x)$ converges absolutely & uniformly on $[c-r, c+r]$.
3. **Preserve R:** $\sum_{n=0}^{\infty} a_n(x-c)^n$ and $\sum_{n=1}^{\infty} n a_n(x-c)^{n-1}$ have same R . ps: 即求导 R 不变
4. **Termwise differentiation:** $f(x)$ is differentiable on $(c-R, c+R)$ || ¹ $f'(x) = \sum_{n=1}^{\infty} n a_n(x-c)^{n-1}, x \in (c-R, c+R)$
² $\forall 0 < r < R, f'(x)$ converges absolutely and uniformly on $[c-r, c+r]$. ps: 即可以对里面求导
5. **Coefficients & Diff:** ¹ $f(x)$ is infinitely differentiable on $(c-R, c+R)$ || ² $a_k = \frac{f^{(k)}(c)}{k!}$
6. **Uniqueness:** If $\sum a_n(x-c)^n = \sum b_n(x-c)^n$ for $\forall x \in \mathbb{R}, |x-c| < r_0 \Rightarrow a_n = b_n$

5.2 Exponential Function & Real Analytic Function

Exponential Function: $E(x) := \sum_{n=0}^{\infty} \frac{x^n}{n!}$ with $R = \infty$ ps: $E(x) = \lim_{n \rightarrow \infty} (1 + \frac{x}{n})^n$ Hint: 证 pointwise 考虑 $\ln$, as $\ln(1 + \frac{x}{n}) = \int_1^{1+\frac{x}{n}} \frac{1}{t} dt$, 缩放, squeeze theorem, uniformly 考虑 unif cont

Real Analytic Function: $f : I \rightarrow \mathbb{R}$ is real analytic iff $\exists (a_n)_{n \in \mathbb{N}} \text{ s.t. } f(x) = \sum_{n=0}^{\infty} a_n(x-c)^n$ for $\forall x \in I$ i.e. Series converges pointwise on I .

Taylor Series: Given ¹Open Interval I , ²infinitely differentiable $f : I \rightarrow \mathbb{R}$. $\Rightarrow$ Taylor Series of f at $c \in I := \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x-c)^n$

· Not all infinitely differentiable functions are real analytic: e.g. $f(x) = \{e^{-1/x^2}, x > 0; 0, x < 0\}$

Taylor's Theorem: If $f : (a, b) \rightarrow \mathbb{R}$ is differentiable function at $x_0 \in (a, b)$:

1. $P_{n, f, x_0}(x) := f(x_0) + \sum_{k=1}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k$ (Taylor's polynomial of degree n at x_0)
2. If $f^{(n+1)}$ exists on (a, b) , then: $\exists \xi \in (x, x_0) \text{ s.t. } f(x) = P_{n, f, x_0}(x) + \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-x_0)^{n+1}$
3. Remainder $R_{n, f, x_0} := f(x) - P_{n, f, x_0}$. || $f : I \rightarrow \mathbb{R}$ is real analytic iff $\lim R_{n, f, x_0} = 0$ for $\forall x \in I$

6 Open & Closed Sets

Open Set: Set E is open iff $\forall x \in E, \exists r > 0, s.t. (x - r, x + r) \subseteq E$

1. **$\cup$ of open sets:** (E_n) are open sets (Infinite). $\Rightarrow \cup E_n$ is open.
2. **$\cap$ of open sets:** (E_n) are open sets (Finite). $\Rightarrow \cap E_n$ is open.
3. **Find Open Set:**
If $f: \mathbb{R} \rightarrow \mathbb{R}$ continuous $\Rightarrow \{x \in \mathbb{R} : f(x) > \lambda \text{ (or } < \lambda)\}$ is open.
4. **Structure of open sets:** $\Downarrow^\ominus$ can be disjoint Hint: $I'_k = \cup I_n \setminus \cup_{i=1}^k I'_i$
 E is a nonempty open set $\Leftrightarrow E = \cup I_n, I_n$ is open interval. (Unique)

Closed Set: Set $F \subseteq \mathbb{R}$ is closed iff $\mathbb{R} \setminus F$ is open.

1. **$\cup$ of closed sets:** (F_n) are closed sets (Finite). $\Rightarrow \cup F_n$ is closed.
2. **$\cap$ of closed sets:** (F_n) are closed sets (Infinite). $\Rightarrow \cap F_n$ is closed.
3. **Find Closed Set:**
If $f: \mathbb{R} \rightarrow \mathbb{R}$ continuous $\Rightarrow \{x \in \mathbb{R} : f(x) \leq \lambda \text{ (or } \geq \lambda) \text{ (or } = \lambda)\}$ is closed.
4. **Sequentially Closed:**
If F is closed set, $\Leftrightarrow \forall (x_n)_{n \in \mathbb{N}} \in F$ convergent. $x = \lim x_n \in F$

Open Set $\rightarrow \cup$ Closed Set: Nonempty open set $E \rightarrow$ union of *disjoint closed* intervals. Hint: ¹Structure of open set. ² each open interval $(a, b) = \cup[a + \frac{1}{n}, b - \frac{1}{n}]$. ³ closed set $\rightarrow$ disjoint.

$\oplus$ Unbounded $\rightarrow \cup$ Bounded: Any closed / open set $F \rightarrow$ countable union of *bounded* closed / open sets. (Hint: $K_n = F \cap [-n, n]$ $K_n = F \cap (-n, n)$)

Both Open & Closed: Only $\emptyset$ and $\mathbb{R}$ are both open and closed.

7 Lebesgue Outer Measure || Lebesgue Measurable || Lebesgue Measure

Countably Additivity: $(E_n)_{n \in \mathbb{N}}$ disjoint sets, m be measure. $\Rightarrow$ Countably Additive if: $m(\cup_n E_n) = \sum m(E_n)$

7.1 Lebesgue Outer Measure

Lebesgue Outer Measure (m^*): Can interpret m^* as $m^*: \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty)$ ps: 本质是寻找到最小的区间可以覆盖 E , 用前面 λ 的定义来定义“长度”

1. **Version 1|Open Intervals:** $m^*(E) := \inf\{\sum_{n=1}^\infty \lambda(I_n) \mid E \subseteq \cup_{n=1}^\infty I_n, I_n \text{ is a open interval}\}$
2. **Version 2|Open Sets:** $m^*(E) = \inf\{\lambda(U), E \subseteq U, U \text{ is open set}\}$ ps: $\lambda(U) = \sum \lambda(I_n)$
3. **Remark:** If we cannot find $(I_n)_{n \in \mathbb{N}}$ s.t. $\sum \lambda(I_n) < \infty \Rightarrow m^*(E) = \infty$.

• **Basic 3 Properties:** Let Set E, F s.t. $E \subseteq F \subseteq \mathbb{R}$.

1. **Null Empty Set:** $m^*(\emptyset) = 0$. || Null Set $E: m^*(E) = 0$.
2. **Monotonicity:** $m^*(E) \leq m^*(F)$.
3. **Countable Subadditivity:** $m^*(\cup_{n=1}^\infty E_n) \leq \sum_{n=1}^\infty m^*(E_n)$.

Useful Properties|By Def of inf: Let E be a set, $m^*(E) < \infty$

Then $\forall \varepsilon > 0, \forall \delta > 0: \exists (I_n)_{n \in \mathbb{N}}, I_n$ open intervals s.t.: $\oplus$

¹ $E \subseteq \cup I_n$ | ² $m^*(E) \leq \sum \lambda(I_n) < m^*(E) + \varepsilon$ | ³ $I_n \cap E = \emptyset, \forall n$ | ⁴ $\lambda(I_n) < \delta, \forall n$

⁵ $I_n \cap E \not\subseteq I_k, \forall n \neq k$. $\leftarrow$ 无覆盖项 无无关项 $\uparrow$ ps: 也使用与是 open set 的.

• **More Properties:**

1. **Interval:** If $J \subseteq \mathbb{R}$ is interval. $m^*(J) = \lambda(J)$.
2. **Countable:** Any Countable set $E \Rightarrow m^*(E) = 0$ $\exists m^*(E) = 0, E$ uncountable (e.g. Cantor Set (Closed Set))
3. $\oplus$ **Further| $m^{**}(E): m^*(E) = m^{**}(E) := \inf\{\sum \lambda(I_n) \mid E \subseteq \cup I_n, I_n \text{ interval}\}$** || $\oplus m^*(E) \leq \sum \lambda(I_n)$, if $E = \cup I_n, I_n$ intervals.
4. **Translation Invariance:** $m^*(E) = m^*(x + E)$

Def of dist(E, F): $dist(E, F) := \inf\{|x - y| : x \in E, y \in F\}$ || **Property:** ¹ E, F are nonempty closed sets; ² $E \cap F = \emptyset$; ³ E bounded. $\Rightarrow dist(E, F) > 0$

Countable Additivity|Outer Measure:

1. **Basic:** If $dist(E, F) > 0 \Rightarrow m^*(E \cup F) = m^*(E) + m^*(F)$.
2. **Closed|Open|Interval Sets:** If $(K_n)_{n \in \mathbb{N}}$ is disjoint, (Closed (bdd) | Open | Interval), sets $\Rightarrow m^*(\cup_{n=1}^\infty K_n) = \sum_{n=1}^\infty m^*(K_n)$

7.2 Lebesgue Measurable

Lebesgue Measurable:

1. **Version 1:** A set $E \subseteq \mathbb{R}$ is *Lebesgue measurable* if $\forall \varepsilon > 0, \exists$ open set $U \subseteq \mathbb{R}$ s.t. $E \subseteq U$ and $m^*(U \setminus E) < \varepsilon$ ps: 被 open set 从外“恰好”包住
2. **Version 2:** A set $E \subseteq \mathbb{R}$ is *Lebesgue measurable* if $\forall \varepsilon > 0, \exists$ closed set $U \subseteq \mathbb{R}$ s.t. $U \subseteq E$ and $m^*(E \setminus U) < \varepsilon$ ps: 被 closed set 从内“恰好”包住

• $\mathcal{M} :=$ collection of all Lebesgue measurable sets of $\mathbb{R}$.

Property of Lebesgue Measurable Sets:

1. **Closed Set|Open Set|Interval:** Any closed set / open set / interval is Lebesgue measurable.
2. **Union $\cup$ | $\oplus$ Intersection $\cap$:** If $(E_n)_{n \in \mathbb{N}}$ is *Lebesgue Measurable* $\Rightarrow$ Also: ¹ $E := \cup_{n=1}^\infty E_n$ $\oplus$ ² $E := \cap_{n=1}^\infty E_n$. Hint: $(\cap E_n)^C = \cup E_n^C$
3. **Complement E^C :** if $E \subseteq \mathbb{R}$ is *Lebesgue Measurable* $\Rightarrow \mathbb{R} \setminus E$ is also *Lebesgue Measurable*.
4. **Measure Zero Set:** $m^*(E) = 0 \Leftrightarrow E$ Lebesgue measurable.
5. **Translation Invariance:** $m(E) = m(x + E)$ || If E is Lebesgue Measurable $\Rightarrow x + E$ also.
6. **Find Lebesgue Measurable Set:** A simple function $s \Rightarrow \{x \in \mathbb{R}, s(x) > \lambda (< \lambda \text{ or } = \lambda)\}$ is Lebesgue Measurable.

7.3 Lebesgue Measure

Lebesgue Measure: *Lebesgue Measure* is a function $m: \mathcal{M} \rightarrow [0, \infty)$ s.t. $m(E) := m^*(E), \forall m \in \mathcal{M}$ ps: 原来是 $m^*: \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty)$, 现在是 $m: \mathcal{M} \rightarrow [0, \infty)$

• **Property: $\star$ All properties of Lebesgue Outer Measure hold for Lebesgue Measure. $\star$**

Countable Additivity for Measure: If $(E_n)_{n \in \mathbb{N}}$ disjoint & *Lebesgue Measurable*. $\Rightarrow m(\cup_{n=1}^\infty E_n) = \sum_{n=1}^\infty m(E_n)$

Monotone Convergence for Measure: If $(E_n)_{n \in \mathbb{N}}$ *Lebesgue Measurable*.

1. **Increasing $\uparrow$:** If $E_n \subseteq E_{n+1} \Rightarrow m(\cup_{n=1}^\infty E_n) = \lim_{n \rightarrow \infty} m(E_n)$
2. $\oplus$ **decreasing $\downarrow$:** If $E_{n+1} \subseteq E_n, m(E_1) < \infty \Rightarrow m(\cap_{n=1}^\infty E_n) = \lim_{n \rightarrow \infty} m(E_n)$ (Hint: $F_n := E_n \setminus E, E = \cap E_n; m(F_n \cup E) = m(E_n)$)

Vitali Set: ¹ 存在 not Lebesgue measurable set. ² 存在 not countability additive set.

• **Example:** $\forall$ cosets $B \in \{x + \mathbb{Q} \mid x \in \mathbb{R}\}$, choose a number $x_B \in B \cap [0, 1]$ $A := \{x_B : x_B \in B\}$ ¹ A is not Lebesgue Measurable Set. ² $E := \cup_{q \in \mathbb{Q} \cap [-1, 1]} (q + A)$ not Countability Additive.

8 Lebesgue Integral

8.1 Step Function || Characteristic Function || Definition of Integral

Characteristic Function: $\chi_E(x) := \{1, x \in E ; 0, x \notin E ; E \subset \mathbb{R}$

Basic Integral Definitions: $\int_{\mathbb{R}} \chi_I(x) dx = m(I) = \lambda(I) \quad || \quad \int_{\mathbb{R}} c \chi_I(x) dx = c \lambda(I) \quad || \quad \int_{\mathbb{R}} \sum c_j \chi_{I_j}(x) dx = \sum c_j \lambda(I_j), c_j \geq 0$

Step Function: $f : \mathbb{R} \rightarrow [0, \infty)$ is a (non-negative) step function if $f(x) = \sum c_j \chi_{I_j}(x)$ for some c_j and χ_{I_j}

· **Property:** If f is a step function $\Rightarrow \exists c_n, I_n$ s.t. I_n disjoint and $f(x) = \sum c_n \chi_{I_n}(x)$

8.2 Simple Functions || Measurable Functions

8.2.1 Simple Functions & Measurable Function

Simple Function: $s : \mathbb{R} \rightarrow \mathbb{R}$ is a simple function iff: $s(x) = \sum_{j=1}^k c_j \chi_{E_j}, E_j$ are Lebesgue Measurable sets. $|| E_j$ can be disjoint.

Integral of Simple Function|Positive: $\int_{\mathbb{R}} s(x) dx := \sum_{j=1}^k c_j m(E_j) \quad || \quad \int_{\mathbb{R}} \alpha s_1 + \beta s_2 = \alpha \int_{\mathbb{R}} s_1 + \beta \int_{\mathbb{R}} s_2$

Measurable Function: $f : \mathbb{R} \rightarrow \mathbb{R}$ is Lebesgue Measurable if $\forall \lambda \in \mathbb{R}, \{x \in \mathbb{R} : f(x) > \lambda\}$ (Superlevel Set) is Lebesgue Measurable.

1. **MF|Simple|Continuous Function:** (Simple|Continuous) Function is Lebesgue Measurable. $\downarrow \text{Hint: } \{x | f(x) < \lambda\}^C = \{x | f(x) > \lambda - \frac{1}{n}\}$

2. **MF|Monotone:** All Monotone functions are Lebesgue Measurable. $\text{Hint: Prove it by using definition directly.}$

3. **MF|Measurable Set:** If f is Lebesgue Measurable, $\Rightarrow \{x \in \mathbb{R} : f(x) > \lambda (< \lambda \text{ or } = \lambda)\}$ is Lebesgue Measurable.

4. **MF|property:** If f, g, f_n is Lebesgue Measurable, $\Rightarrow \max\{f, g\}, \min\{f, g\}, f \cdot \chi_E, \alpha f \pm \beta g, f^2, f \cdot g, \sup f_n, \inf f_n, \limsup f_n, \liminf f_n$ also.

5. **Measurable Func $\Rightarrow \exists$ Simple Func:** f is 非负勒贝格可测函数 $\Rightarrow \exists$ simple functions $(s_n)_{n \in \mathbb{N}}, 0 \leq s_n \leq s_{n+1}(\uparrow)$ s.t. $s_n \rightarrow f$ pointwise.

8.2.2 Measurable Functions|Positive

Integral of Measurable Function|Positive: Let $f : \mathbb{R} \rightarrow [0, \infty)$ be Lebesgue Measurable Function.

1. **Basic Def:** $\int_{\mathbb{R}} f(x) dx := \sup \{ \int_{\mathbb{R}} s(x) dx : s \text{ simple}, 0 \leq s \leq f \}$

2. **Extend Def:** If $f : \mathbb{R} \rightarrow \mathbb{R}$ Lebesgue Measurable, E Measurable Set. $\Rightarrow \int_E f(x) dx := \int_{\mathbb{R}} f(x) \chi_E(x) dx$

Basic Properties of Integral|Positive: Let $f, g : [0, \infty) \rightarrow \mathbb{R}$ be non-negative Lebesgue Measurable functions.

1. **Monotonicity:** If $0 \leq f(x) \leq g(x) \Rightarrow 0 \leq \int_{\mathbb{R}} f(x) dx \leq \int_{\mathbb{R}} g(x) dx$

2. **Markov's Inequality:** For all $\lambda > 0 \Rightarrow m(\{x \in \mathbb{R} : f(x) > \lambda\}) \leq \frac{1}{\lambda} \int_{\mathbb{R}} f(x) dx$ 函数积分面积 > 其对应底下矩形面积

3. **Measure Zero:** If E s.t. $m(E) = 0 \Rightarrow \int_E f(x) dx = 0$ $\text{Hint: } \forall \text{ simple } s, 0 \leq \int_E s = \int_{\mathbb{R}} s \chi_E = \int_{\mathbb{R}} c_n \chi_{E_n \cap E} = \sum c_n m(E_n \cap E) \leq \sum \dots$

MCT for Integral|Positive: $^1 f_n : [0, \infty) \rightarrow \mathbb{R}$ non-decreasing ($\uparrow$) MF; $^2 \lim f_n = f$ pointwise. $\Rightarrow ^1 f$ is measurable; $^2 \int_{\mathbb{R}} \lim f_n dx = \lim \int_{\mathbb{R}} f_n(x) dx$

More Properties of Integral|Positive: Let $f, g : [0, \infty) \rightarrow \mathbb{R}$ be non-negative Lebesgue Measurable functions.

1. **Linear Combination:** $\alpha, \beta > 0 \Rightarrow \int_{\mathbb{R}} \alpha f + \beta g = \alpha \int_{\mathbb{R}} f + \beta \int_{\mathbb{R}} g$

2. **Union of Disjoint:** E, F disjoint measurable sets $\Rightarrow \int_{E \cup F} f = \int_E f + \int_F f$

3. **'MCT for Int Series':** $^1 f_n$ non-negative MF; $^2 \sum f_n = f$ pointwise $\Rightarrow ^1 f$ is measurable; $^2 \int_{\mathbb{R}} \sum f_n = \sum \int_{\mathbb{R}} f_n$

4. **(Reverse) Fatou Lemma:** f_n non-negative MF $\Rightarrow \int_{\mathbb{R}} \liminf f_n \leq \liminf \int_{\mathbb{R}} f_n \leq (\limsup \int_{\mathbb{R}} f_n \leq \int_{\mathbb{R}} \limsup f_n)$

8.3 Integrating Real-Valued Functions

Isolated Functions: For a function f . We define that $f^+ := \max\{f, 0\}, f^- := -\min\{f, 0\} \Rightarrow f = f^+ - f^-, |f| = f^+ + f^-$.

· **Remark:** If $a < 0: (af)^+ = (-a)f^-$ and $(af)^- = -af^+$

Lebesgue Integrable: A MF $f : \mathbb{R} \rightarrow \mathbb{R}$ is Lebesgue Integrable iff $\int_{\mathbb{R}} |f(x)| dx < \infty \Leftrightarrow \int_{\mathbb{R}} f^+ dx < \infty \& \int_{\mathbb{R}} f^- dx < \infty$

1. **Def of Integral:** If $f \in \mathcal{L}^1(\mathbb{R}) \Rightarrow \int_{\mathbb{R}} f(x) dx := \int_{\mathbb{R}} f^+(x) dx - \int_{\mathbb{R}} f^-(x) dx$

2. **Extended Def:** If $f : E \rightarrow \mathbb{R}$ with E Measurable Set s.t. $\tilde{f} := f(x) \chi_E$ is Lebesgue Integrable $\Rightarrow \int_E f(x) dx := \int_{\mathbb{R}} \tilde{f}(x) dx$

3. $\mathcal{L}^1(\mathbb{R}) :=$ Collection of all Lebesgue Integrable functions on $\mathbb{R}$.

4. $\mathcal{L}^1(E) :=$ Collection of all Lebesgue Integrable functions on E .

Property of Integral: Let $f, g \in \mathcal{L}^1(\mathbb{R})$. Then:

1. **Triangular Inequality for Integrals:** $|\int_{\mathbb{R}} f(x) dx| \leq \int_{\mathbb{R}} |f(x)| dx$

2. **Linearity of the Integral:** $\alpha f + \beta g \in \mathcal{L}^1(\mathbb{R})$ and $\int_{\mathbb{R}} \alpha f + \beta g = \alpha \int_{\mathbb{R}} f + \beta \int_{\mathbb{R}} g \Rightarrow \mathcal{L}^1(\mathbb{R})$ is a vector space

3. **'Comparison Test':** If $^1 f : \mathbb{R} \rightarrow \mathbb{R}$ measurable $^2 \int_{\mathbb{R}} g(x) < \infty$ $^3 |f| < g \Rightarrow f \in \mathcal{L}^1(\mathbb{R})$

4. **Remark:** Can use p-test, Comparison test ... when we want to prove a function is Lebesgue Integrable.

Dominated Convergence Theorem (DCT): $^1 f_n$ is MF. $^2 f_n \rightarrow f$ pointwise. $^3 \exists g \in \mathcal{L}^1(\mathbb{R})$ s.t. $|f_n| \leq g \Rightarrow ^1 f \in \mathcal{L}^1(\mathbb{R})$ & $\lim \int_{\mathbb{R}} f_n = \int_{\mathbb{R}} f$

· **Uniform Convergence Theorem:** If $^1 \lim f_n(x) = f(x)$ uniformly on $E \in \mathcal{M}, m(E) < \infty, ^2 f_n \in \mathcal{L}^1(\mathbb{R}) \Rightarrow ^1 f \in \mathcal{L}^1(E)$ $^2 \lim \int_E f_n(x) dx = \int_E f dx$

Fundamental Theorem of Calculus (FTC): $f : [a, b] \rightarrow \mathbb{R}$ be continuous. $\Rightarrow ^1 F(x) := \int_a^x f(y) dy$ is differentiable $^2 F'(x) = f(x)$

· **Corollary:** $f : [a, b] \rightarrow \mathbb{R}$ be continuously differentiable. $\Rightarrow \int_a^b f'(x) dx = f(b) - f(a)$

· **Remark:** $^1 f$ is Lebesgue Integrable. (By Extreme value theorem) and. 2 需要闭区间 (Closed) 否则需要结合 MCT 或 DCT 极限逼近来用.

1. **Integration-By-Parts:** If $f, g : [a, b] \rightarrow \mathbb{R}$ be continuously differentiable. $\Rightarrow \int_a^b f g' dx = [f g]_a^b - \int_a^b f' g dx$

2. **Integration-By-Substitution:** If $f : [c, d] \rightarrow \mathbb{R}$ continuous & $\phi : [a, b] \rightarrow [c, d]$ continuous diff $\Rightarrow \int_a^b f(\phi) \phi' dx = \int_{\phi(a)}^{\phi(b)} f(y) dy$

9 L^p Spaces || Hölder’s Inequality || Minkowski’s Inequality

10 Appendix

Inequation:

Triangular	$ x + y \leq x + y $	$ x - y \geq x - y $	$ x - y \leq x \pm y .$
AM-GM & Cauchy	$a + b \geq 2\sqrt{ab}, a > 0, b > 0$	$a^2 + b^2 \geq 2ab$	$(\sum a_nb_n)^2 \leq \sum a_n^2 \sum b_n^2$
Bernoulli’s Inequality	$(1 + \delta)^\alpha \leq 1 + \delta\alpha, \alpha \in (0, 1], \delta \geq -1$		$(1 + \delta)^\alpha \geq 1 + \delta\alpha, \alpha \in [1, +\infty), \delta \geq -1$
$e^x, \ln(x)$	$e^x \geq 1 + x \geq x$ Hint: Def/MVT	$\ln x \leq x - 1 \leq x$ Hint: MVT	$e^x \geq 1 + x + \frac{x^2}{2} + \cdots$ Hint: Def $\frac{1}{k} \leq \ln(x) - \ln(x - 1)$ Hint: $\int_{k-1}^k \frac{dx}{x}$
常见变换技巧	$(1 + \frac{x}{n})^n \geq 1 + n \cdot \frac{x}{n} + C_n^2(\frac{x}{n})^2$ Hint: Binomial Theorem		$ a_n x - c ^n = a_n \rho^n \cdot \frac{ x-c ^n}{\rho^n}, x - c < \rho < R$
$\sin(x), \cos(x)$	$ \sin(x) \leq x $ Hint: MVT	$1 - \cos(x) < x $ Hint: MVT	
常见变换技巧 2	If $\forall \varepsilon > 0 \ a \leq b + \varepsilon \Rightarrow a \leq b$	$\forall \delta, 0 < \delta < 1$ and $\delta a \leq b \Rightarrow a \leq b$	

和差化积/积化和差: s+s=sc,s-s=cs,c+c=cc,c-c=-ss || e.g. $\sin A + \sin B = 2\sin(\frac{A+B}{2})\cos(\frac{A-B}{2})$ $\sin A \sin B = -\frac{1}{2}[\cos(A+B) - \cos(A-B)]$ Hint: 和差角公式相加减

$\sin(2x) = 2\sin(x)\cos(x)$ $\cos(2x) = 2\cos^2(x) - 1 = 1 - \sin^2(x) = \cos^2(x) - \sin^2(x)$ $\cos(x) = \sqrt{\frac{1+\cos(2x)}{2}}$ $\sin(x) = \sqrt{\frac{1-\cos(2x)}{2}}$

Hyperbolic Func: $\sinh(x) = \frac{e^x - e^{-x}}{2}$ $\cosh(x) = \frac{e^x + e^{-x}}{2}$ $\cosh^2(x) - \sinh^2(x) = 1$ $[\sinh(x)]' = \cosh(x)$ $[\cosh(x)]' = \sinh(x)$
 $\sinh(x \pm y) = \sinh(x) \cosh(y) \pm \cosh(x) \sinh(y)$ $\cosh(x \pm y) = \cosh(x) \cosh(y) \pm \sinh(x) \sinh(y)$

Common Int: $\int \tan(x)dx = -\ln|\cos(x)|$; $\int \cot(x)dx = \ln|\sin(x)|$; $\int \sec(x)dx = \ln|\sec(x) + \tan(x)|$; $\int \csc(x)dx = \ln|\csc(x) - \cot(x)| = \ln|\tan(\frac{x}{2})|$

Common Der: $[\tan(x)]' = \sec^2(x)$; $[\cot(x)]' = -\csc^2(x)$; $[\sec(x)]' = \sec(x)\tan(x)$; $[\csc(x)]' = -\csc(x)\cot(x)$; $[\arctan(x)]' = \frac{1}{1+x^2}$.
 $1 + \tan^2(x) = \sec^2(x)$; $1 - \csc^2(x) = \cot^2(x)$; $\csc^2(x) - \cot^2(x) = 1$. $[\arcsin(x)]' = \frac{1}{\sqrt{1-x^2}}$; $[\arccos(x)]' = -\frac{1}{\sqrt{1-x^2}}$

Interchanging the Order of Summation: $\sum_{n=1}^p \sum_{m=1}^q f(n, m) = \sum_{m=1}^q \sum_{n=1}^p f(n, m)$ || $\sum_{k=0}^n \sum_{i=0}^k = \sum_{i=0}^k \sum_{k=i}^n$ Hint: 列出来看 or 画图

Summation By Parts: $\sum_{k=1}^n a_k(b_{k+1} - b_k) = (a_{n+1}b_{n+1} - a_1b_1) - \sum_{k=1}^n (a_{k+1} - a_k)b_{k+1}, \forall (a_k)_{k=1}^{n+1}, (b_k)_{k=1}^{n+1}$
 $\Rightarrow \sum_{k=1}^n a_k b_k = a_{n+1}\sigma_{n+1} - \sum_{k=1}^n (a_{k+1} - a_k)\sigma_{k+1}, \forall (a_k)_{k=1}^{n+1}, (b_k)_{k=1}^{n+1}$ $\sigma_k := \sum_{j=1}^{k-1} b_j$

q^n - 1: $q^n - 1 = (q - 1)(1 + q + q^2 + \cdots + q^{n-1})$ || $1 \pm \mathbf{x}^3: 1 \pm x^3 = (1 \pm x)(1 \mp x + x^2)$

Useful Counterexample: $(1/2)^{\frac{1}{n}} \subseteq [1/2, 1)$

Common Power Series: $\sum_{n=0}^\infty x^n = (1 - x)^{-1}$ $\sum_{n=1}^\infty nx^{n-1} = (1 - x)^{-2}$ $\sum_{n=2}^\infty n(n - 1)x^{n-2} = 2(1 - x)^{-3}$ $R = 1$ for all.

$\sum_{n=0}^\infty \frac{x^n}{n!} = e^x$ $\sum_{n=1}^\infty (-1)^{n+1} \frac{x^n}{n} = \ln(x + 1)$ $\sum_{n=1}^\infty \frac{x^n}{n} = -\ln(1 - x)$ $R = \infty$ for all.

Sum of n, n^2, n^3: $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ $\sum_{k=1}^n k^3 = [\frac{n(n+1)}{2}]^2$

Countably Approximation Tools: $\sum_{n=1}^\infty 2^{-n}\varepsilon = \varepsilon$

De Morgan’s Laws: $(A \cap B)^C = A^C \cup B^C$ $(A \cup B)^C = A^C \cap B^C$

$E \setminus F = F^C \cap E$ is open (i.e. open - closed = open) || $F \setminus E = E^C \cap F$ is open (i.e. closed - open = closed)

特殊集合构造技巧: $\{x \in \mathbb{R} : (f + g)(x) > \lambda\} = \bigcup_{t \in \mathbb{Q}} (\{x \in \mathbb{R} : f(x) > \lambda - t\} \cap \{x \in \mathbb{R} : g(x) > t\})$ || $\{x \in \mathbb{R} : \forall \varepsilon > 0, \exists N > 0 \ s.t. \ \dots < \varepsilon\} = \bigcap_{k=1}^\infty \bigcup_{N=1}^\infty \{x \in \mathbb{R} : \dots < \frac{1}{k}\}$